

DURVA DEV

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Education

The Pennsylvania State University – University Park

08/2024 – 05/2026

Master of Science in Computer Science

University of Mumbai, India

08/2018 – 05/2022

Bachelor of Engineering in Computer Engineering

Experience

Agentic AI Developer @ Pennsylvania State University

State College, PA

Python, AWS Strands Agents, Typescript, Playwright

01/2026–05/2026

- **Member of global top-10 team for the Amazon NOVA Challenge** Built agents on AWS for automated code generation with Claude Code; Evaluated using OpenTelemetry, Prometheus, and SigNoz.

Systems Engineer @ Tata Consultancy Services

Mumbai, India

Python, React, SQL, Linux, Mainframe, Bash, IBM MQ

07/2022 – 07/2024

- Built an enterprise Gen AI solution for TCS BaNCS RFP automation over a 3K+ document enterprise knowledge base using RAG; adopted by sales teams and reduced response turnaround from 3–4 weeks to 2–3 days.
- Developed document ingestion, chunking, embedding, and retrieval pipelines with metadata-aware search and prompt engineering, defined retrieval metrics and reduced search latency by 40%.
- Delivered 8+ projects for Saudi National Bank across customer lifecycle, including regulatory government compliance features, delinquent loan monitoring, and enhancement of fee-processing platforms handling 100K+ daily transactions.
- Developed and integrated IBM MQ-based asynchronous reporting workflows with third-party vendors, and debugged Oracle PL/SQL stored procedures on databases containing 11M+ rows using execution plan analysis to diagnose and resolve production failures.

Hackathons

DoomScholar: #1 in c0mpiled-6 hackathon. Built an **anti-doomscrolling iOS app** that detects passive scrolling and interrupts with AI-generated quizzes from LMS material using SwiftUI, WKWebView scroll tracking, FastAPI, and OpenAI API, deployed on Vercel.

Projects

TutorTrace | *React, TypeScript, Dexie.js, IndexedDB, OAuth2, Supabase*

- Built an offline-first Canvas LMS study app with AI flashcard generation from course content, a gamification engine with XP and streaks, Canvas OAuth2, and non-blocking full-text search via a Web Worker.

LedgerLink: Personal Finance App | *React, FastAPI, PostgreSQL, Plaid, Docker, MCP*

- Developed an AI-powered financial assistant utilizing an agent interacting with an MCP-enabled data store to automate expense tracking, enabling users to analyze synced Plaid banking data through natural language queries, and generate proactive insights.

OWASP Security RAG Agent | *Python, ChromaDB, AWS Bedrock, Strands*

- Built an agentic RAG pipeline over OWASP WSTG and Exploit-DB/GHDB; parsed and embedded 18K+ documents into ChromaDB vector stores; agent dynamically routes queries across both knowledge bases and generates grounded answers via Claude Haiku on Amazon Bedrock.

Locksmith | *Go, PostgreSQL, GCP, Terraform, Kubernetes, GitHub Actions*

- Built a fault-tolerant job queue in **Go** and **PostgreSQL** using `FOR UPDATE SKIP LOCKED` for concurrent worker claiming; implemented exponential backoff with jitter, dead-letter queues, priority aging, and lease recovery

Linearizable KV store | *Go, RPC*

- Built a fault-tolerant key-value store in Go using Raft consensus; achieved **65K reads/sec** (p99 1.0ms) and **5.8K writes/sec** (p99 11.5ms) with sub-20ms failover and crash-safe durability via append-only WAL and periodic snapshots.

MapReduce Framework | *C++, gRPC, Protobuf, AWS, HDFS*

- Engineered a MapReduce framework in C++ with gRPC inter-node communication; processed 100MB+ datasets under 3s across EC2 instances with 64 concurrent threads; implemented thread-safe nodes with HDFS partitioning and tracing to eliminate data races.

Technical Skills

Languages. C/C++, Go, Javascript/Typescript, Python (Pytorch, Tensorflow)

CI/CD & Automation. Docker, GitHub Actions, Kubernetes, Prometheus, Grafana, OTel, Terraform

Cloud & Infrastructure. AWS EC2, S3, Lambda, DynamoDB, Bedrock, CodeBuild, IAM, CloudWatch, VPC Azure OpenAI, AI Search GCP GKE, Cloud SQL, Artifact Registry, Secret Manager, VPC, Vertex AI

Databases & Storage. PostgreSQL, Oracle SQL, Hadoop File System, Redis, ChromaDB

AI. LangGraph, CrewAI, LangChain, Google ADK, MCP, OpenAI API, Pydantic